ECO 630 Course Project: A Micro-Costing Analysis of Alcohol Screening and Brief Intervention (ASBI) in Primary Care: Practitioner vs. Specialist Delivery Models

INTRODUCTION

Excessive alcohol use in North Carolina represents a significant public health concern, imposing over $9.72 billion in annual costs to society (Gora Combs et al., 2022, p. 217). A majority of these costs (77%) arise specifically from binge drinking, defined as four or more drinks for women or five or more drinks for men on a single occasion (CDC, 2024). The disproportionate burden created by binge drinking in NC, however, is not reflected in who ultimately bears the costs, with 45.6% paid by the government, only 38.7% paid by the drinker, and 15.7% paid by other parties (Gora Combs et al., 2022, p. 217). The significance of these negative externalities created by excessive alcohol use highlights the need for strategies that aim to minimize harmful consumption behaviors and mitigate costs to society.

Alcohol screening and brief intervention (ASBI) is an evidence-based strategy that has demonstrated effectiveness in reducing excessive alcohol use. A systematic review of 37 adult trials found that brief intervention strategies reduce the number of drinks per week, the percentage of individuals that exceed recommended drinking limits, and episodes of binge drinking (Curry et al., 2018, p. 1906). Furthermore, research shows that "those with higher consumption at baseline tend to experience a larger effect" (Beyer et al., 2019), indicating that individuals who are the most at risk and generate a disproportionate share of alcohol-related costs stand to benefit the most from intervention. Proactive prevention strategies with proven efficacy offer a viable alternative to relying solely on costly downstream responses to alcohol-related harm.

The multiple implementation strategies for ASBI necessitate a thorough cost-effectiveness analysis to identify the approach that maximizes benefit while ensuring an efficient allocation of resources. ASBI evaluations may be administered by existing medical personnel or by a dedicated staff of non-medical individuals who specialize in intervention strategies. Due to the varied resource inputs and program costs inherent in each of these options, a cost-effectiveness analysis is required to help guide efficient allocation and identify the approach that achieves the greatest reduction in harmful alcohol use per dollar spent.

METHODS

Cost Analysis

This analysis is conducted from the healthcare system perspective. Included costs reflect direct program expenditures and staff time associated with screening and brief intervention activities within participating primary care clinics. Patient-incurred costs and broader societal impacts were not included.

Wage data for primary care providers, nurse practitioners, primary care nurses, and health educators were provided by the key informant. Annual work hours were

assumed to equal 2,080 hours (52 weeks x 40 hours) and fringe benefits were set at 29.5% based on the latest report from the Bureau of Labor Statistics (2025). Provider-delivered BIs are assumed to integrate into existing workflows in the P-model. In the S-model, a health educator is assumed to support the full ASBI workflow, including scoring the Health Assessment Survey for all eligible (non-follow-up) visits, scoring of the Cutting Back screen for HAS-positive patients, delivery of brief interventions, and follow-up encounters. Because the educator is modeled as a dedicated program resource, educator labor costs are calculated as 80% of annual compensation divided by annual eligible (non–follow-up) visit volume per clinic.

According to labor wage data and information provided by the key informant, provider training for ASBI delivery under the P-model requires one hour of provider time and $60 per trainee, which includes the printing costs of training materials. Training costs were allocated across one year of eligible visit volume.

Guilford County reports that 18% of adults engage in binge or heavy drinking with a margin of error of 15–22% (County Health Rankings, 2025). In the absence of system-specific screening positivity data, this prevalence was used to approximate the proportion of screened patients likely to meet criteria for brief intervention. This estimate is consistent with the prevalence of at-risk drinking observed in primary care clinics in Babor et al. (2005), which ranged from 8.0–15.0% across implementation sites (p. 365).

According to information from the key informant, the system of 25 clinics has, on average, 6 primary care providers that see 20 patients a day of which, half are follow-up visits. These values were used to estimate a daily screening volume of 1,500 patients across the network. Assuming the binge drinking rate of 18% from above, this translates to 270 patients per day who receive an intervention. As the clinics operate five days a week, the expected annual patient volume is 780,000, of which 390,000 are not follow-up visits, with 70,200 (18%) receiving an initial BI. It is expected that half of patients receiving a BI will receive a follow-up BI lasting up to 5 minutes.

Materials related to the ASBI program consist of the Health Assessment Survey form, Cutting Back Survey, and educational pamphlet. Both the HAS form and Cutting Back Survey are single-page B&W and result in $0.05 each in printing costs. The HAS form is provided to all eligible patients, while the Cutting Back Survey is administered only to patients who screen positive on the HAS. The educational pamphlet costs $1.35 to produce and is provided to every patient receiving a BI. Additionally, half of patients who receive a follow-up BI request a second copy of the pamphlet.

Health outcomes

Reduction in alcohol consumption was measured by average drinks per week as an indicator of changes in drinking habits following an ASBI. Estimates were obtained from Babor et al. (2006), which reported baseline and three-month follow-up consumption levels for control, provider, and specialist-delivered interventions. At baseline, participants reported an average of 10.2 drinks per week in both the control and provider groups and 10.6 drinks per week in the specialist group. At the three-month follow-up, average consumption declined to 8.2 drinks per week in the control group, 6.6 drinks per week in the provider group, and 6.7 drinks per week in the specialist group. These reductions show that both of the ASBI delivery models produced larger reductions in short-term alcohol consumption than the control condition.

Reduction in the frequency of binge drinking was also examined as a measure of ASBI's effect on harmful alcohol use. Curry et al. (2003) report the proportion of participants engaging in binge drinking at the three-month follow-up, with 22 percent of individuals in the control group reporting binge drinking compared with 13 percent among those receiving the intervention. Further research is required to distinguish outcomes between the provider-delivered and specialist-delivered ASBI models. As such, these reductions were not included in CEA tables but nonetheless provide supporting evidence that screening and brief intervention can reduce harmful drinking behaviors.

Estimates of health-related quality of life gains were sourced from Barbosa et al. (2015). The authors estimate QALYs gained by comparing baseline and follow-up health utility values following ASBI implementation. The intervention generated gains of 0.013 QALYs in emergency department settings and 0.008 QALYs in outpatient settings relative to baseline health states. As this research also does not specify QALY gains based on the provider-delivered and specialist-delivered ASBI models, an average of these values was used to approximate the expected quality of life benefit of ASBI, resulting in an estimated 0.010 QALYs gained per patient.

Resources saved

Reductions in emergency department (ED) visits were estimated using results reported by Fleming et al. (2002). The study reports an average of 0.98 ED visits per patient in the control group (n = 382; 376 total visits) and 0.77 visits per patient in the intervention group (n = 392; 302 total visits), a reduction of 0.21 visits per patient. The valuation of this reduction was developed using the average cost of a treat-and-release ED visit and was obtained from the Agency for Healthcare Research and Quality. As the intervention targets adult alcohol use, a weighted average cost was calculated using ED visits for patients aged 18 and older, yielding an estimated $820.89 per visit (Table 1, AHRQ, 2024). Applying this value to the observed reduction of 0.21 ED visits per patient results in an estimated $172.39 in healthcare resources saved per patient associated with the intervention.

Incremental Cost-Effectiveness

Cost-effectiveness analysis (CEA) was used to compare the costs and outcomes associated with differing delivery methods for alcohol screening and brief intervention by

examining the additional cost required to reduce alcohol consumption by an additional unit. The incremental cost-effectiveness ratio (ICER) captures this relationship by calculating the difference in costs between two alternatives divided by the difference in their outcomes. In the short-term analysis, this can be read as the extra cost required to reduce alcohol consumption by one additional drink per week. Interventions are ordered from least to most costly and compared to determine incremental costs and effects. If an intervention is both less costly and more effective than another, it is considered dominant, and the dominated alternative is assumed to be undesirable to the decision-maker.

Sensitivity Analysis

A one-way deterministic sensitivity analysis was conducted to examine how changes in labor costs affect the results. Specifically, the specialist delivering BI in the S-model was assumed to receive the same annual compensation and fringe benefits as the provider delivering the BI in the P-model. The specialist program delivery cost was then recalculated using the same annualization approach applied in the base case, while all other assumptions, including shared material costs, health outcomes, and estimated resources saved, were held constant. The short-term and long-term CEAs were then recalculated using the revised specialist labor cost.

RESULTS

Cost Analysis

Table 1 summarizes the estimated cost per eligible visit for implementing alcohol screening and brief intervention (ASBI) under the provider-delivered and specialist-delivered approaches. Because shared program costs apply equally to both strategies, differences between delivery methods arise primarily from labor inputs required to administer screening and intervention activities.

Under the P-model, these tasks are performed by existing clinical staff, including nurses and primary care providers, resulting in a total estimated cost of $4.202 per eligible visit. On the other hand, the S-model employs a dedicated health educator to perform most program activities. Despite introducing a dedicated health educator role, the lower wage rate of the educator results in a lower total cost of $3.684 per eligible visit. Overall, the specialist-delivered approach is approximately $0.52 cheaper per eligible visit than the provider-delivered model.

Incremental Cost-Effectiveness

Table 2A presents the short-term CEA using reductions in average drinks per week. Both intervention models produced larger reductions in alcohol consumption than the control condition. Because program costs are driven primarily by staff labor time, the specialist-delivered model achieves similar reductions in alcohol consumption at a lower cost than the provider-delivered model. As a result, the specialist-delivered intervention dominates the provider-delivered approach in the short-term analysis.

Table 2B presents the long-term analysis using quality-adjusted life years (QALYs) gained as the outcome. Both intervention models generate an estimated gain of

0.010 QALYs per patient relative to the control condition. After including cost savings that arise from avoided emergency department visits, both interventions result in lower net costs than the control condition. Because the specialist-delivered model has the lowest net cost while producing the same health benefit as the provider model, it dominates both the provider-delivered intervention and the control condition.

Sensitivity Analysis

The sensitivity analysis shows that the base-case findings rely heavily on the lower wage for the specialist-delivered model. When the specialist's labor is valued at the provider wage, the specialist program costs increase from $3.321 to $13.282 per eligible visit. This raises the total S-model cost from $3.684 to $13.645, while the P-model remains at $4.202. In the short-term analysis, the provider-delivered model becomes more cost-effective than the specialist-delivered model under the revised wage assumption. The specialist-delivered model requires much higher spending to achieve only a small additional reduction in drinks per week. In the long-term analysis, both intervention models continue to generate the same estimated QALY gain, but the higher specialist cost raises the net cost of the S-model above that of the P-model. Under this assumption, the provider-delivered intervention dominates the specialist-delivered intervention, while the control condition remains dominated.

CONCLUSION

Excessive alcohol consumption creates a significant public health and economic burden in North Carolina. Most of these costs are driven by binge drinking, while much of the burden is carried by parties other than the individual drinker. This creates a strong policy incentive for upstream strategies that reduce harmful alcohol use before it results in more serious health and economic consequences. ASBI is one potential strategy, but its value depends both on its ability to reduce harmful drinking and how it is implemented. This analysis aimed to evaluate the effectiveness of ASBI programs delivered in primary care settings by comparing provider-delivered and specialist-delivered models to identify the approach offering the most efficient use of healthcare resources.

The results of this evaluation show that ASBI is a promising intervention strategy under either delivery model. Both approaches produced larger reductions in alcohol consumption than the control condition alone and positive long-term health gains relative to no intervention. In the base-case analysis, the specialist-delivered model was preferred because it achieved similar outcomes at a lower cost than the provider-delivered model. However, the sensitivity analysis showed that this conclusion depends heavily on specialist labor costs. When specialist wages were assumed to equal provider wages, the provider-delivered model became the more cost-effective option. These findings suggest that healthcare systems should support ASBI implementation in primary care while selecting the delivery model that best fits their staffing structure and labor costs.

Several limitations should be considered when interpreting these results. First, the analysis was conducted from the healthcare system perspective and did not include patient-incurred costs, societal costs associated with excessive alcohol use, or social

benefits associated with reductions. Second, the proportion of patients expected to screen positive was approximated using county-level excessive drinking prevalence rather than system-specific screening data. Third, the estimates of QALY gains and emergency department visits avoided were extrapolated from prior studies and were not specific to the region or delivery models examined here. In addition, the specialist-delivered model relied on key informant assumptions regarding staffing, workflow, and visit volume, while the sensitivity analysis examined only one change in labor cost. These limitations suggest that the magnitude of the cost-effectiveness estimates should be interpreted with caution.

Although these limitations should be kept in mind, the overall findings remain promising. The analysis suggests that reducing harmful alcohol use through primary care intervention can be an efficient use of healthcare resources under a wide range of implementation assumptions. Even under the sensitivity analysis, where the specialist was assumed to earn the same wage as a primary care provider, the net per-patient cost of intervention remained negative relative to the control condition. Furthermore, these estimates exclude broader downstream effects on social productivity, family stability, accident rates, and incarceration, suggesting that the benefits of ASBI outlined here are likely understated. Given the scale of the burden created by excessive alcohol use, expanding ASBI in primary care is a practical, evidence-based approach to reducing harm and limiting avoidable downstream costs.

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TABLES AND FIGURES

Table 1: Cost of Alcohol Screening and Brief Intervention

Shared Costs			
Activity	Resources used	Unit cost of resource	Total cost per eligible visit
Health Assessment	Single Page B&W	$0.05/page	$0.050
Cutting Back Form	Single Page B&W	$0.05/page	$0.009
Intervention Materials	Educational Pamphlet	$1.35/ea	$0.304
Unique Costs for Provider Delivered Intervention (P-Model)			
Provider Training	Provider Labor (annualized)	$0.071 per eligible visit	$0.071
Assessment Scoring	PCN Labor	$0.726/min	$1.452
Cutting Back Scoring	PCN Labor	$0.726/min	$0.261
Provider BI	Provider Labor	$2.075/min	$1.121
Provider BI Follow-up	Provider Labor	$2.075/min	$0.934
Unique Costs for Specialist Delivered Intervention (S-Model)			
Specialist Program Administration	Specialist Labor (annualized)	$3.321 per eligible visit	$3.321
		Grand Total (P-Model)	$4.202
		Grand Total (S-Model)	$3.684

Table 2A: Short-Term Cost-Effectiveness Analysis

Treatment	Cost	Incremental Cost	Reduction in Drinks/Week	Incremental Effect	ICER ($ per drink reduced)
C (Control)	0.000	—	2.0	—	—
S (Specialist)	3.684	3.684	3.9	1.9	1.94
P (Provider)	4.202	0.518	3.6	-0.3	Dominated

Table 2B: Long-Term Cost-Effectiveness Analysis

Treatment	Net Cost	Incremental Cost	QALYs	Incremental QALYs	ICER ($/QALY)
S (Specialist)	-168.706	—	0.010	—	—
P (Provider)	-168.188	0.518	0.010	0.000	Dominated
C (Control)	0.000	168.188	0.000	-0.010	Dominated

Table 3: ASBI Costs Under Provider-Equivalent Specialist Wages

Shared Costs			
Activity	Resources used	Unit cost of resource	Total cost per eligible visit
Health Assessment	Single Page B&W	$0.05/page	$0.050
Cutting Back Form	Single Page B&W	$0.05/page	$0.009
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Unique Costs for Provider Delivered Intervention (P-Model)			
Provider Training	Provider Labor (annualized)	$0.071 per eligible visit	$0.071
Assessment Scoring	PCN Labor	$0.726/min	$1.452
Cutting Back Scoring	PCN Labor	$0.726/min	$0.261
Provider BI	Provider Labor	$2.075/min	$1.121
Provider BI Follow-up	Provider Labor	$2.075/min	$0.934
Unique Costs for Specialist Delivered Intervention (S-Model)			
Specialist Program Administration	Specialist Labor at Provider Wage (annualized)	$13.282 per eligible visit	$13.282
		Grand Total (P-Model)	$4.202
		Grand Total (S-Model)	$13.645

Table 4A: ASBI Short-Term Sensitivity Analysis

Treatment	Cost	Incremental Cost	Reduction in Drinks/Week	Incremental Effect	ICER ($ per drink reduced)
C (Control)	0.000	—	2.0	—	—
P (Provider)	4.202	4.202	3.6	1.6	2.63
S (Specialist)	13.645	9.443	3.9	0.3	31.48

Table 4B: ASBI Long-Term Sensitivity Analysis

Treatment	Net Cost	Incremental Cost	QALYs	Incremental QALYs	ICER ($/QALY)
P (Provider)	-168.188	—	0.010	—	—
S (Specialist)	-158.745	9.443	0.010	0.000	Dominated
C (Control)	0.000	158.745	0.000	-0.010	Dominated